

Probabilistic Modeling of the Blood Microbiome in CRC

Generative Metagenomic Analysis via Dirichlet Multinomial Mixtures (DMM) for Liquid Biopsy

Colorectal Cancer (CRC) Project

1. Biological Context

Colorectal cancer (CRC) compromises the integrity of the intestinal mucosal barrier, promoting the translocation of bacterial DNA into the bloodstream.

- **Bacterial Load:** Cancer patients exhibit a high blood 16S rRNA gene load (mean: 7,628 copies/ μ L colon, 9,145 copies/ μ L rectum).
- **High Risk:** Patients $> 95^{\text{th}}$ percentile ($> 13,000$ copies/ μ L) have an **OR of 17.08** for colon cancer.
- **Permeability and Diet:** The combination of elevated zonulin and bacterial load increases risk (OR = 4.55). Garlic reduces risk by 73%.

2. Metagenomic Challenge

The analysis of circulating bacterial DNA presents complexities that invalidate traditional Euclidean clustering methods (e.g., K-Means):

- **Sparsity & High Dimensionality:** Thousands of OTUs dominated by the *Rare Biosphere* (low-abundance taxa).
- **Uneven Sampling:** Variable sequencing depth (53 – 10,585 reads/sample), generating high stochastic noise.
- **Discrete Data:** Counts constrained to the proportions simplex.

3. The DMM (Dirichlet Multinomial Mixtures) Generative Framework

To overcome metric limitations, the DMM generative Bayesian model (Holmes et al.) is adopted, which evaluates the generative probability of profiles instead of Euclidean distance:

$$\text{Bayesian Formulation: } P(\Theta|X) \propto P(X|\Theta) \cdot P(\Theta) \quad | \quad \text{Multinomial Likelihood: } L_i(X_i|\pi_i) = J_i! \prod_{j=1}^S \frac{\pi_{ij}^{X_{ij}}}{X_{ij}!}$$

- **Dirichlet Prior:** The parameter α is decomposed into a mean vector m and a concentration factor h ($\alpha = m \cdot h$) to model metacommunity variance on the simplex.
- **EM & BFGS Algorithm:** The marginalized evidence is optimized via the Expectation-Maximization cycle and the quasi-Newtonian BFGS algorithm.
- **Selection Criterion (K):** Choice of the number of clusters K based on Laplace Approximation and Hessian matrix (Occam's Razor).

4. Clinical Results and Diagnostic Prospects (Liquid Biopsy)

By pre-filtering taxa with < 10 counts and discretizing clinical variables, the DMM analysis highlights partial biological correspondence:

Cluster Separation:

Cluster 1 isolates healthy subjects, whereas Cluster 2 groups CRC patients along with a subset of healthy individuals (Chi-Square Test $p < 0.05$).

Discriminant Taxa:

Differential variation for *Bacteroides*, *Prevotella*, *Roseburia*, *Faecalibacterium*, and *Ruminococcus*.

- **Anna Karenina Principle:** Healthy blood ecosystems are uniform and predictable, whereas tumor dysbiosis presents structural chaos with high inter-individual variance.
- **Prospects:** The combination of metagenomic sequencing and DMM lays the foundation for non-invasive precision screening via liquid biopsy.

References

- [1] Holmes, I., Harris, K., & Quince, C. (2012). Dirichlet Multinomial Mixtures: Generative Models for Microbial Metagenomics. *PLoS ONE*, 7(2), e30126. <https://doi.org/10.1371/journal.pone.0030126>